

Recommending Makerspace Best Practices Based On Visualization of Student Use Data

8th International Symposium on Academic Makerspaces

ISAM
2024
Paper
No.:
18

J.Phan¹, J. Truong²

¹J. Phan; Frank Batten School of Leadership and Public Policy, University of Virginia; e-mail: jordan.phan@virginia.edu

²J. Truong; College of Arts and Sciences., University of Virginia; e-mail: hcz3yb@virginia.edu

Abstract

The University of Virginia Scholar's Lab Makerspace offers a wide variety of tools and resources for Charlottesville community members to make and fabricate. The Makerspace collects data through two avenues, recording information regarding users and their visits as well as equipment reservation and use. Usage of the Makerspace peaks in the early afternoon, and centers around the middle of the month, when the Makerspace tends to hold special events, and equipment tends to be used most often on weekdays and towards the beginning and end of the semester, indicating that the Makerspace has both recreational and practical appeal. Users tend to be affiliated with either the College of Arts and Sciences or the School of Engineering and are often drawn to activities and tools with low learning curves and barriers to entry. We recommend that (1) the Makerspace implement changes to increase consistency in data collection for better analysis, (2) that Technologists schedule hours with regard to highest use times, and that (3) the Makerspace implement specific strategies to encourage additional visits.

Introduction

The UVA Scholar's Lab Makerspace is a space for students, staff, faculty, and community members to learn and make collaboratively. The Makerspace is staffed by students via two avenues: students can either be paid Technologists, or work as volunteers for limited hours. The Makerspace is open over 40 hours a week, and offers 3d printing services, sewing and textiles, soldering and electronics, digital art, miscellaneous crafting supplies, and more. The Makerspace has supported hundreds of users through workshops, training, and technical support, including interdisciplinary projects with UVA medical and other departments. We are interested in analyzing the use of space and equipment, as well as identifying traits of users who frequent the space. As the Makerspace continues to grow, we will use this analysis to promote its resources to users who may be unaware, as well as to expand services that are in high demand. We aim to make the Makerspace more accessible and helpful to the community we serve.

Previous research on tool usage by mechanical engineering students show that 3D printers and metal tools were among the most popular tool groups offered [1]. Additionally, metal tool use reached a peak with 3rd year students, declining again with additional years of study. In MIT's MakerWorkshop,

Du and Haji found that their most frequently used machines were their 3D printer, laser cutter, and hand tools [2], along with analyzing diversity trends in their student-run space. By examining similar trends in our own space, we can better support our users, increase availability of high-demand tools, and identify gaps in our data.

We aim to answer the following research questions:

1. What are the most common reasons that users visit the Makerspace?
2. How can we increase visits?
3. What are best practices to develop to support students' needs?

Methods

Data were collected in two ways; the first primarily targets visitors to the Makerspace, while the second targets equipment usage. Upon entry or exit, visitors are asked to fill out a form via a small kiosk by the door to record their visit. The form asks for a visitor's name, email, their student, faculty, or staff status, their school or affiliation, and the reason for their visit. Visitors are also offered the opportunity to scan a QR code to sign up for the Makerspace newsletter. The visit data were collected between March of 2022 and March of 2024.

When visitors inform a Technologist that they wish to use Makerspace equipment, the technologist helps the visitor fill out a form to create a reservation, taking down the user's name and email. Visitors may also fill out this form in advance of their visit to reserve equipment ahead of time. This system allows Makerspace staff to set hours of availability for various equipment, track reservations and equipment use, and analyze the dates and times of most frequent use. The equipment data spans from 2014 to 2024, though some data is missing prior to 2023.

Our data collection and analysis are limited in a few distinct ways. Firstly, most visitors to the Makerspace are reflected in at least one of the aforementioned data sets; however, some visitors fail to complete the visit form, and not all equipment requires check-out, so some equipment use and visits are not recorded. As a result, we may be failing to observe key patterns of behavior among those who are less proactive about checking in, or those who use equipment that does not require check-out. Our 3D printers have only been operational for a few months, and our laser cutter is not yet available for public use, so our data is heavily biased against those items.

Secondly, due to the scope of the data, we were unable to perform any predictive or regression analyses. The questions asked on both the visitor and the registration form are not comprehensive, and do not survey users about type of project or detailed personal data. Finally, our visitors were not randomly sampled for survey, and our observations were not set up as a controlled experiment. We are unable to make predictions or establish causal links, but nonetheless we were able to draw conclusions from the data. We developed visualizations using these data sets to make inferences about visitors, visits, and equipment usage.

Results & Discussion

A. MAKERSPACE USER DATA

The Makerspace is open six days a week with varying hours each day, closing only during school-wide holidays. The hours and days of availability change by semester depending on Technologist and volunteer availability. Fig. 1 shows that visits peak at 3:00 p.m., with a gradual increase and sharper decrease before and after that time. There are visits logged at all hours between 10:00 a.m. and 9:00 p.m. While it is rare that regular operating hours for the Makerspace extend past 8:00 p.m., Technologists and volunteers are permitted 24-hour access, so it is possible that some visits logged occur outside of opening hours. Peak visits overall occur around the early afternoon, between 1:00 and 3:00 p.m.

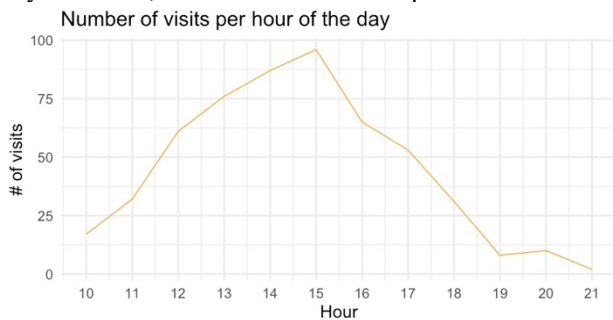


Figure 1: Visits for each hour of the day

We were interested to see if any visitor trends appeared over the course of the month, as shown in Fig. 2. Overall visits peaked on the 9th and the 14th of each month. Each year, the Makerspace holds holiday-themed celebrations, including Valentine's Day, Pi Day, and Mother's Day. During these celebrations, free snacks, easy crafts, and pre-made trinkets are offered, drawing in larger-than-average crowds. Both Pi Day and Valentine's Day are on the 14th of the month, and though the date of Mother's Day varies, it often falls around the 9th and students come in the whole week prior to prepare.

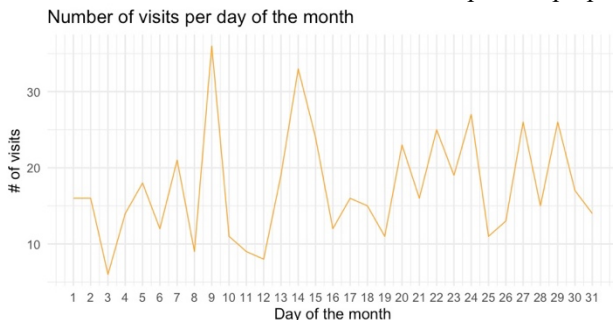


Figure 2: Visits for each day of the month

Figs. 3 and 4 show anonymized users' visits, accumulated over time. Among users that have logged multiple visits, most users visit about 3 times. Patron #151 logged a record 67 visits, and patron #125 logged the second most visits with 13. Patron #151 is co-author Jennifer Truong, who frequented the Makerspace as a user before becoming a volunteer and ultimately a Technologist. Only 31 patrons have more than one visit logged. However, upon manual inspection of the non-anonymized data, Technologists reported that there were several users who visited more often than shown in the graph. Our visit recording system and thus data collection is imperfect – some visitors do not fill out the form unprompted, and Technologists on shift may be occupied or otherwise forget to ask them to sign in. We are therefore likely missing dozens of visits that have occurred. There is potentially a substantial amount of data missed at the busiest, most critical hours. The accuracy aspect of the data is of lower quality and could be more precise.

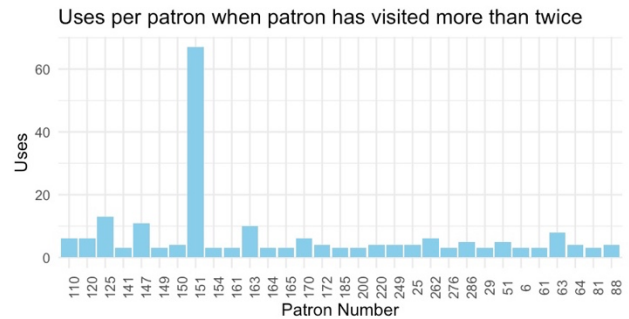


Figure 3: Visits per patron when patron has visited more than twice

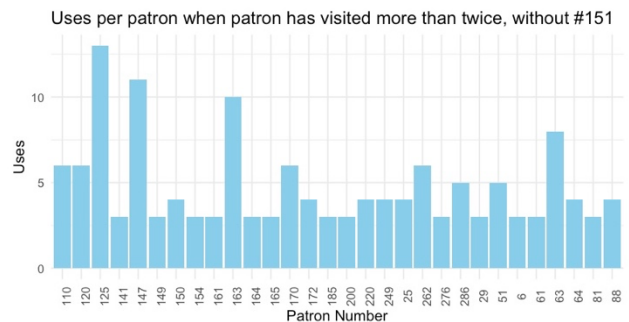


Figure 4: Visits per patron when patron has visited multiple times, without patron #151

Most visitors are affiliated with the College of Arts and Sciences (Fig. 5). At UVA, the College encompasses Sciences, Social Sciences, and Humanities subjects, and these three categories represent three of the four most common visitor affiliations. The second most common affiliation is the School of Engineering & Applied Sciences. The College is the largest school at UVA, and engineering students tend to have the highest awareness of Makerspaces. Architecture students visit the fifth most frequently, likely because they have making needs similar to engineers, but they have their own FabLab hosted at the Architecture school with many resources. Given this information, the Makerspace may choose to conduct outreach and partner with the schools of Commerce, Public Policy, or Nursing, or Education.

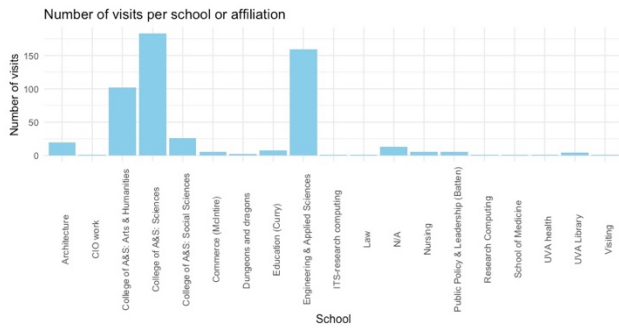


Figure 5: Visits by school or affiliation

The most popular and well-known item by far is the button maker (Fig. 6). Users can come in to make backpack pins, magnets, or keychains, and organization leaders or student candidates often use this free resource to make merchandise. The sewing and textiles station, Silhouette Cameo machine, digital art station, and workshops are also popular reasons to visit. All these tools have low barriers to entry, are not offered elsewhere on campus, and can be used for quick crafting. As a Makerspace that is not tailored to a specific discipline, easily accessible activities are important for boosting visits and creating resources for the whole community. The Makerspace only recently began offering 3D printing services, and follow-up will be necessary to determine its relative popularity over time. However, anecdotally, there is near-daily use of the 3D printers, at rates similar to the button making supplies. Additionally, the Makerspace installed a laser cutter during the Spring 2023 semester, which has not yet become available for public use. We will continue to track use of both pieces of equipment.

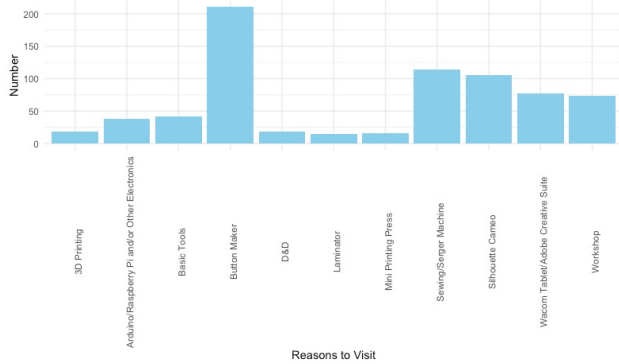


Figure 6: Visit by reason to visit, when more than 10 uses are logged

B. EQUIPMENT USE DATA

The schedule of students in particular can be restrictive, this can limit the opportunities to utilize the makerspace. Visitors use the makerspace both academically and recreationally, spontaneously or scheduled. The nature of some projects requires more hours on certain equipment and may be time-sensitive.

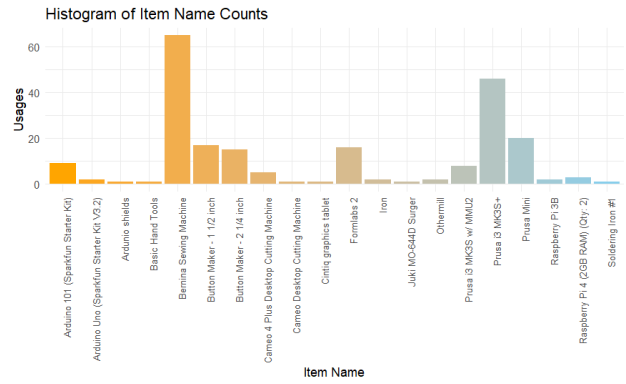


Figure 7: Count of Equipment Bookings

Because of this, observing the trends of when the two most used reservable tools, the sewing machine and 3D printers, are utilized provides useful insights to visitors on what is a good day/time to drop-in or reserve. The button makers have been known to be popular and have been accounted for over their well-established presence and the right quantity of machines are provided for the demand.

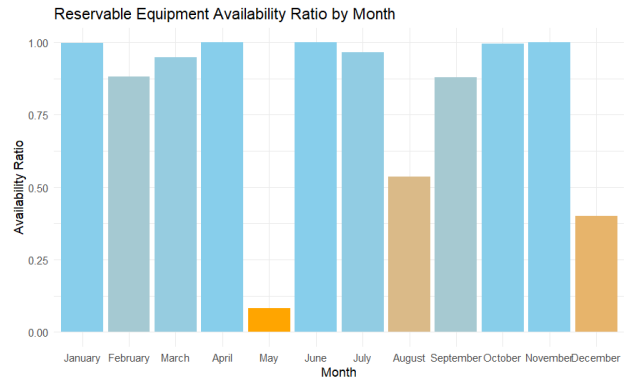


Figure 8: Monthly Availability Ratio

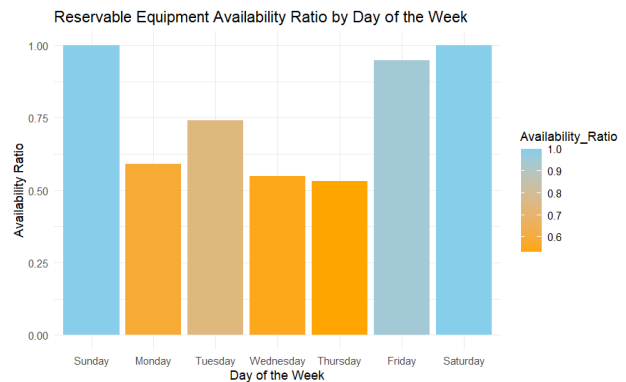


Figure 9: Weekly Availability Ratio

Of the hours opened, the reservable equipment is most heavily booked and utilized during the weekdays and the months towards the start and end of the semesters. A possible explanation is that those months are often when free time is available and/or final projects are due.

Equipment requiring reservations tends to be more specialized equipment that needs a bit of training or guidance by staff, less so for repeat visitors or regulars. Staff presence at hours of frequent reservations would be ideal.

The 3D printers in particular are equipment for which reservations are important. By library policy, the Makerspace

cannot run 3D printers overnight due to fire risk and an abundance of caution for the newly renovated Shannon Library. Cameras are being set up to allow the potential to print overnight in the future. For now, visitors often need to plan their day ahead of time to begin the printing process and when to pick it up. While the hours of operations are up, those change by the semester, and we are interested in how much of the open time is utilized. Those interested in printing may not be able to reserve a printer on short notice on some days and find that a day is fully booked.

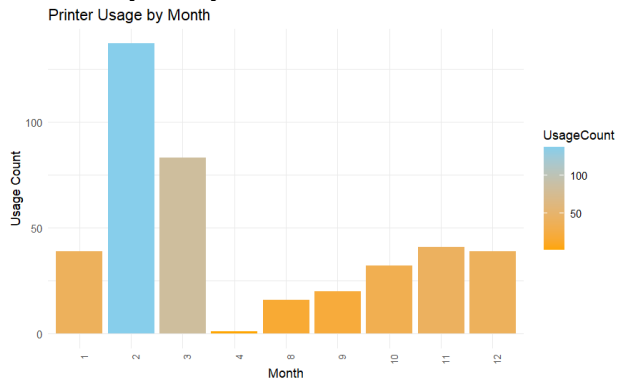


Figure 10: 3D Printer Usage by Month

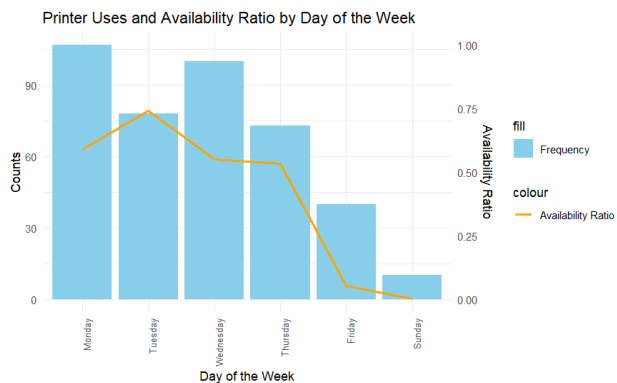


Figure 11: 3D Printer Usage by Day

The trend for 3D printing is that reservations tend to be booked early in the semester and least often in the summer months when students are on campus less, workers and visitors both. There is a slight extremity for the starting months of the semester, August and January, where the semester begins and fewer days available, so the ratio is more responsive to a smaller amount of reservations. The observable pattern in 11 for printers mimics the ones from Figs. 8 and 9 on general reservable equipment usage. To conclude, an ideal time to get printing done would be on weekend days (that the lab is open on) towards the middle of the semester. Further study is needed to determine if these trends hold in the new Makerspace area.

The sewing machines are also used frequently. Hobbyists and students involved in fashion clubs frequently come in for reasons that range from tailoring to creating an entire outfit from scratch. No matter the work, the process is very time-consuming and requires setting aside a block of time.

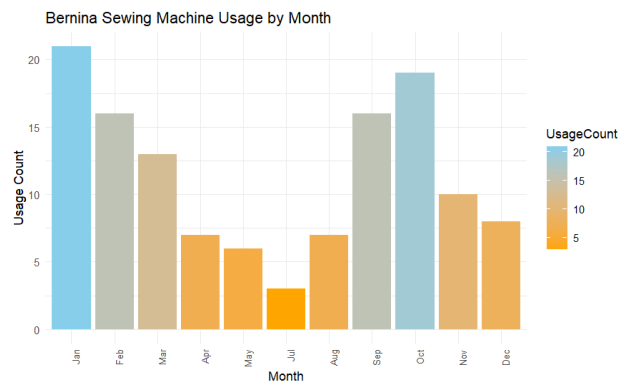


Figure 12: Sewing Machine Usage by Month

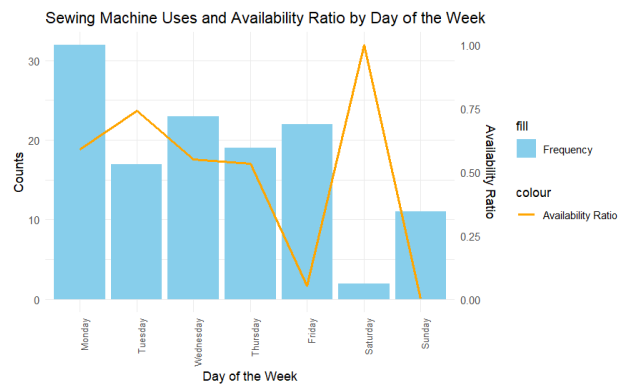


Figure 13: Sewing Machine Usage by Day

The Makerspace has two Bernina sewing machines, making sewing resources more limited in availability than even the 3D printers. With that in mind, the days in which they tend to be free are the weekends and early in the semester. Club events and showcases typically do not occur early in the semester and they are often meant to show off the work done during that school period. This observation is aligned with the data and could be of consideration when collaborating with such organizations. Saturday appears to be quite free and makes for a good day to schedule a technologist skilled in sewing.

Conclusion

Our takeaways and recommendations for best practices follow.

Proactively collect data across all users to address data limitations. Users are generally asked to sign in on the front kiosk, but Technologists often forget or miss users before they can ask. Standardizing check-in procedures, including establishing an expected check-in time at the beginning or end of each visit will create a routine that encourages consistent data collection. Additionally, assigning Technologists to particular roles and areas of the Makerspace, including a check-in role if staff are available, will encourage accountability and consistent attention across all tools. Proactive, comprehensive data collection procedures will help to address the limitations identified in the Methods and allow us to further analyze data to support our users.

Arrange open hours and publication of hours around high-use periods to better serve students. Given that visits tend to peak in the early afternoon (Fig. 1), the Makerspace

should consider scheduling additional Technologists to be available during these hours to provide increased support. Further, publishing usage trends and reservations, particularly on high-use equipment like sewing machines and 3D printers (Figs. 11, 12, 13), may help users to plan visits more efficiently. This information could be published on the Makerspace website, newsletter, or through a live display positioned near the entrance to the Makerspace.

Increase visits through special events and low barrier-to-entry crafts. We found that visits peak during times of the month when the Makerspace hosts special events. Additionally, the button maker, Silhouette Cameo (for sticker making), and Technologist-hosted workshops were among the most common reasons to visit, indicating that users are highly attracted by crafting activities and special events. To increase users and visibility of Makerspace resources, putting on 1-2 special events per month and widely advertising them can draw in new visitors who can be introduced to the Makerspace's work. Crafts with a low learning curve that can be completed easily with a tangible end product like buttons and stickers are also useful to draw visitors. Preserving workshop materials and instructions for quick and easy crafts, as well as advertising their availability, may draw in bigger crowds.

It is important to keep in mind the endorsement of changes is based upon data for reservable equipment and thus made with them in mind. It is possible that setting up a tracking system for non-reservables in the future may provide new insights and recommended actions.

References

- [1] S. Blair et al., "Tool Usage Patterns of Mechanical Engineering Students in Academic Makerspaces," *IEEE Frontiers in Education*, Oct. 2023.
- [2] L. W. Du and M. N. Haji, "Following Diversity in a Student-Run Makerspace: Trends in gender, engagement, and usage," *International Symposium on Academic Makerspaces*, Oct. 2019.